

Phishing URL Detection — Training Report

Dataset: url_features_extracted1
Total usable rows: 100872
Phishing rows: 63492 | Legitimate rows: 37380

Model Comparison

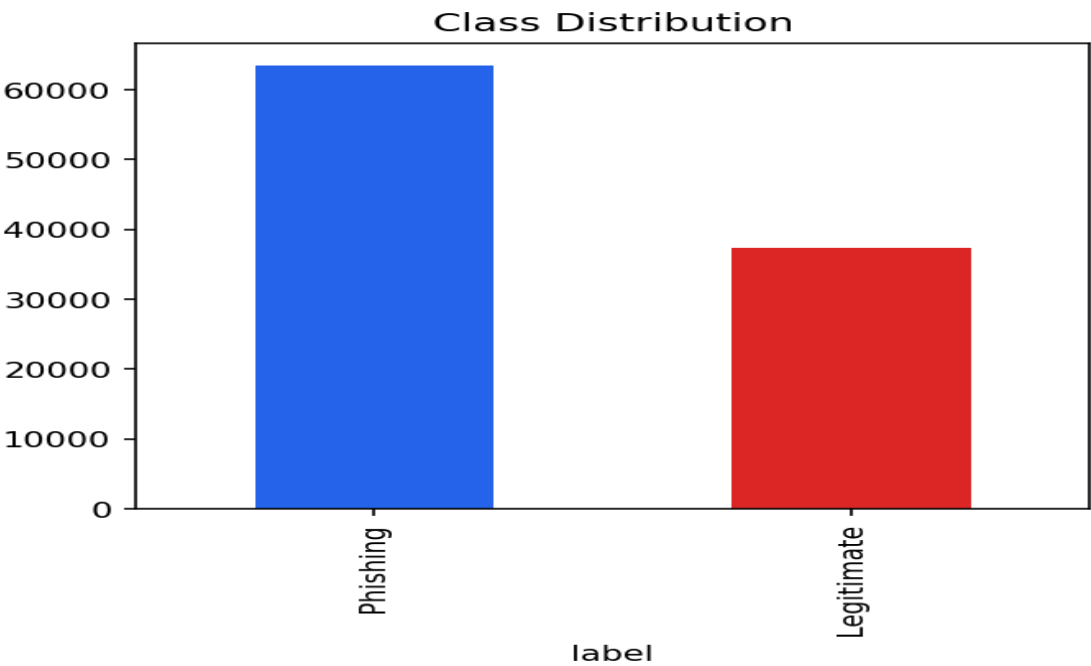
Model	Accuracy	Precision	Recall	F1	ROC-AUC
LogisticRegression	0.9983	0.9996	0.9977	0.9987	0.9999
RandomForest	0.9991	0.9988	0.9999	0.9993	1.0000
ExtraTrees	0.9993	0.9991	0.9998	0.9994	1.0000
GradientBoosting	0.9991	0.9988	0.9998	0.9993	1.0000
AdaBoost	0.9986	0.9986	0.9991	0.9988	1.0000
HistGradientBoosting	0.9995	0.9994	0.9998	0.9996	1.0000

Best model selected: HistGradientBoosting

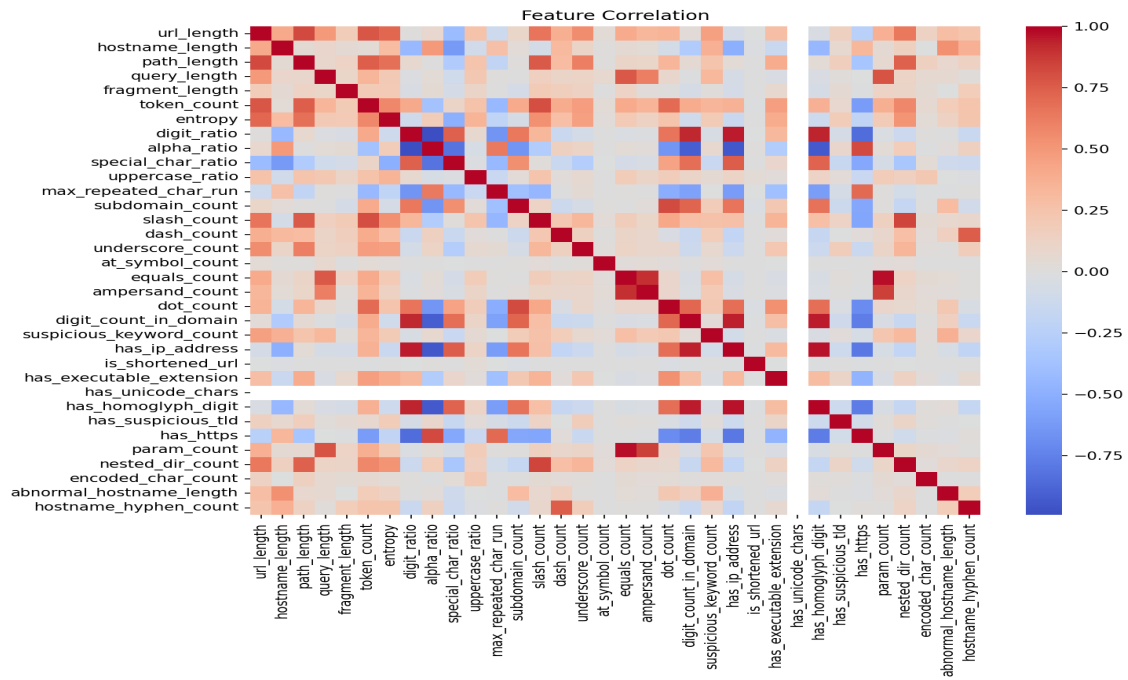
Unseen-Domain Generalization Check

A holdout set was built from 8152 domains never seen in training (16370 rows). Performance on this set: accuracy=0.9993, F1=0.9995, ROC-AUC=0.9998. This demonstrates the model is learning phishing *behavior* rather than memorizing specific domains.

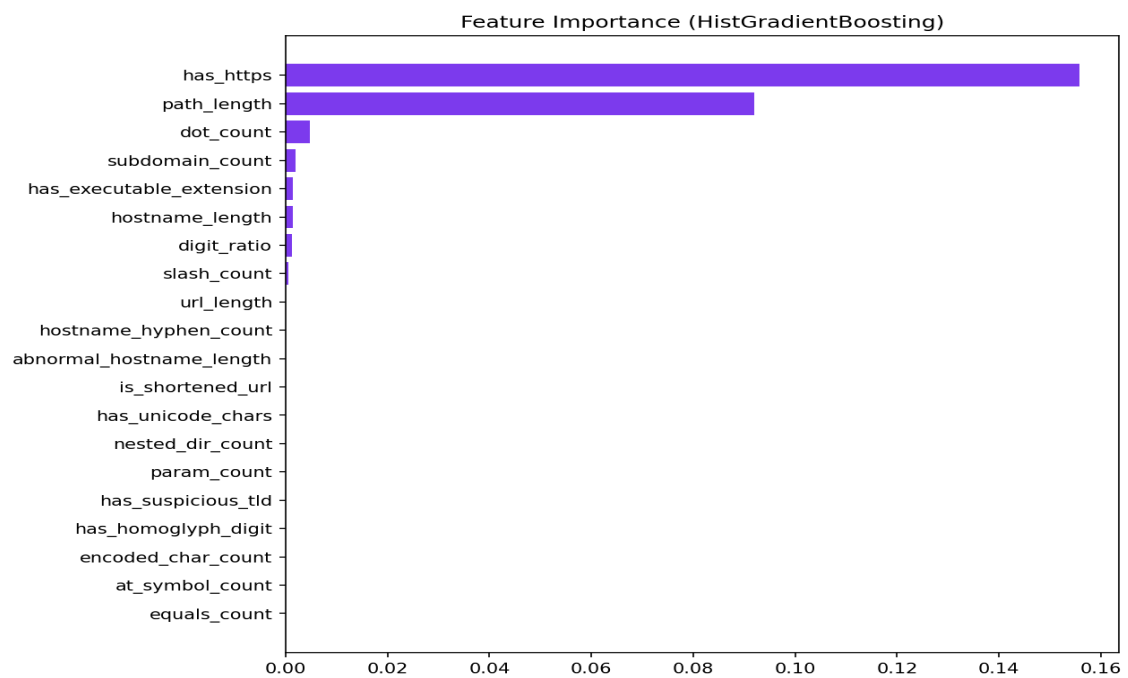
Class Distribution



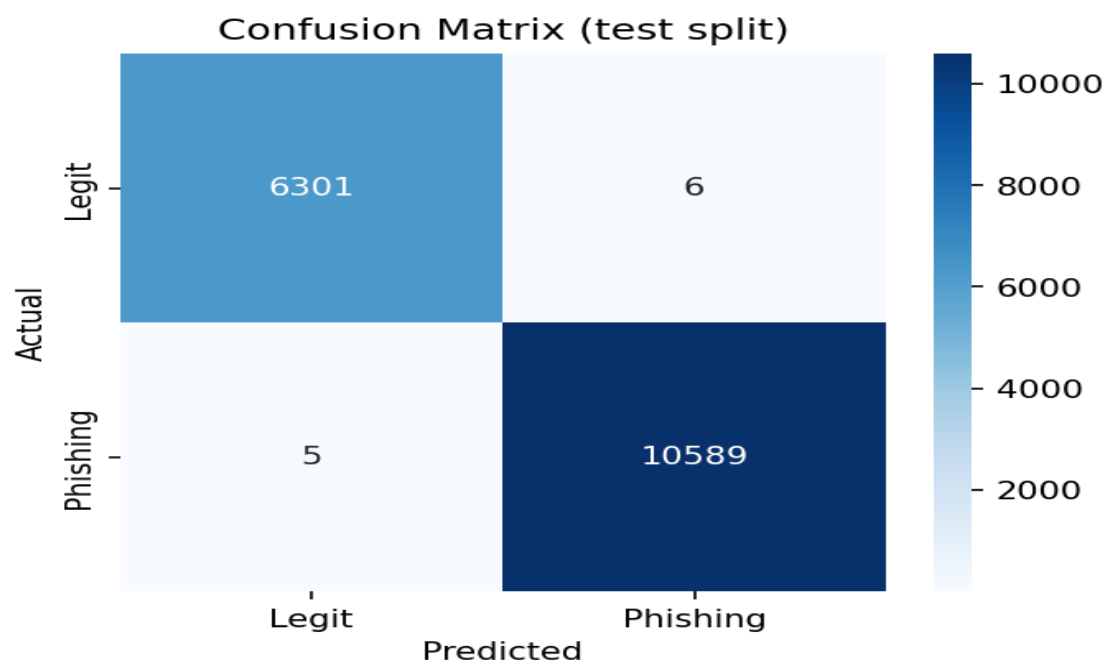
Feature Correlation



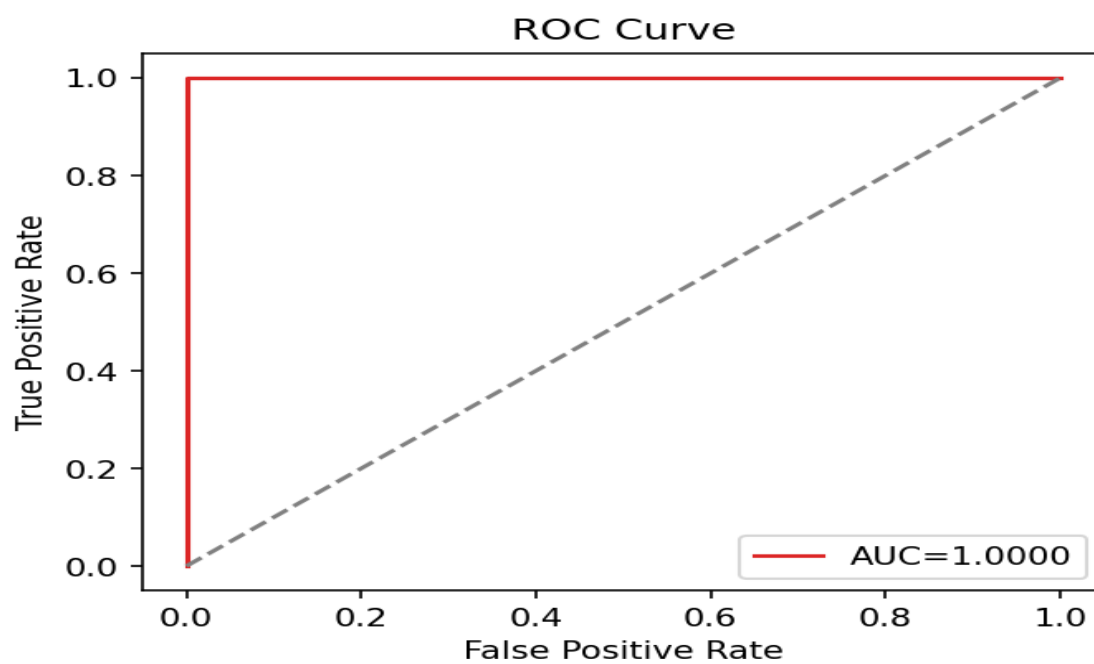
Feature Importance



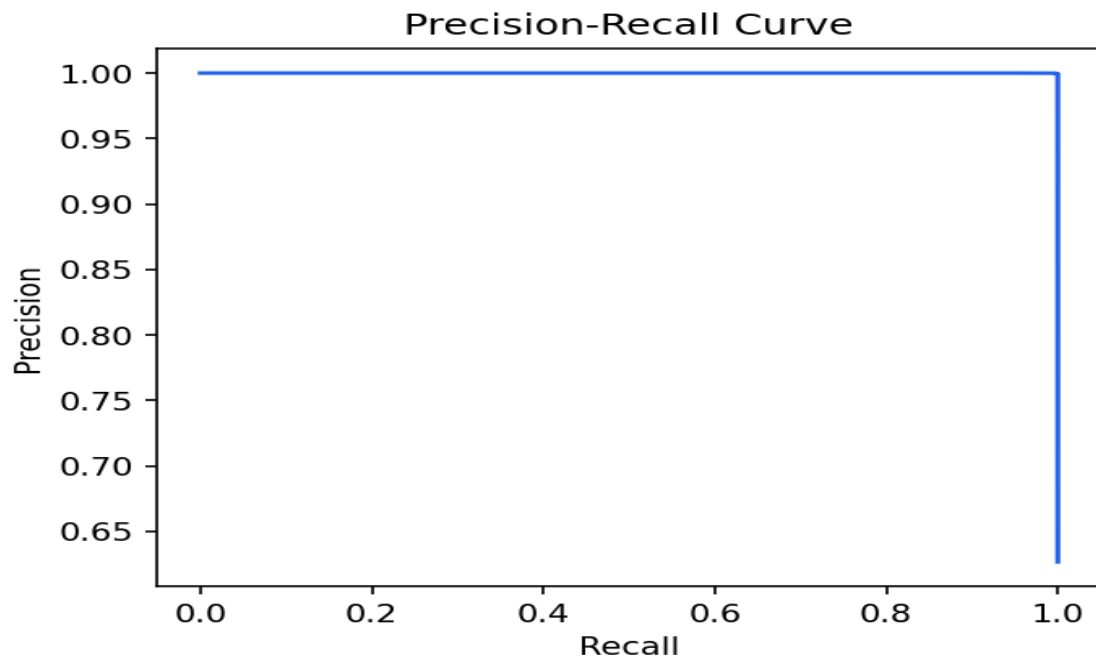
Confusion Matrix



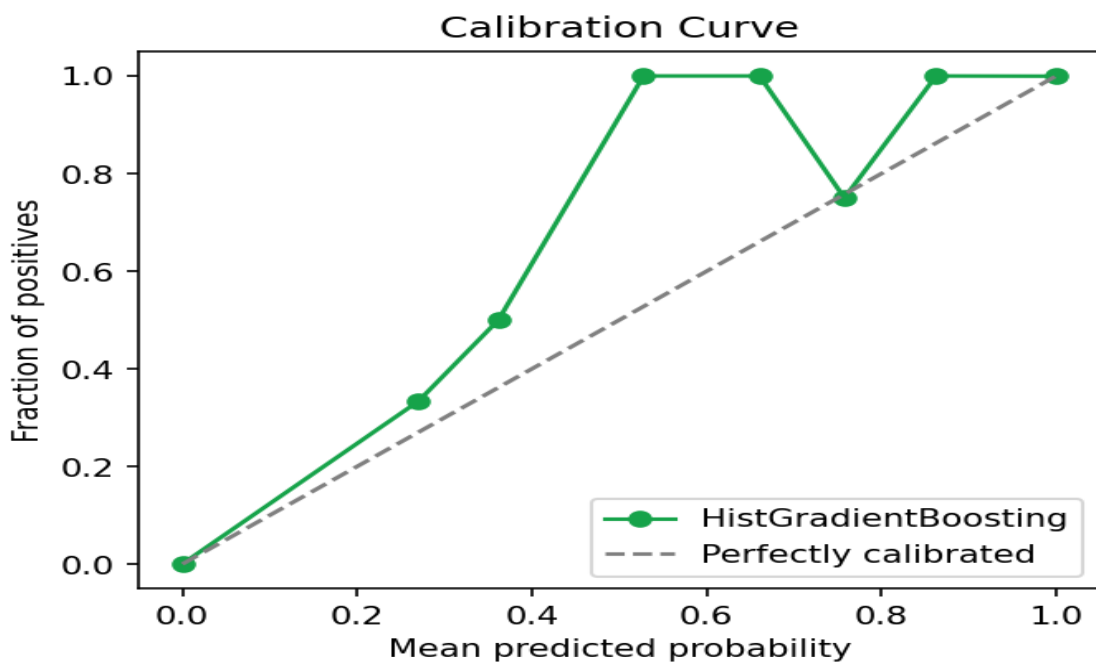
ROC Curve



Precision-Recall Curve



Calibration Curve



Model Comparison (CV F1)

